Using CNNs to identify localised demyelination in a cohort of patients with Multiple Sclerosis using T1 MPRAGE MRI

Lesions of neurodegeneration are characteristic of multiple sclerosis and are detectable using T1 MRI. In this preliminary study we explore the efficacy of a simple CNN in the prediction of lesion masks based on T1 MPRAGE images. This simple CNN model produces predicted lesion masks with a binary accuracy of 93.2% relative to clinician made ground truth masks. Whilst this approach is limited by noisy input data, this study shows that MS lesions are characterisable and detectable using simple CNNs.

Introduction

Multiple sclerosis (MS) is a neurodegenerative inflammatory disease characterised by localisations of demyelination across the central and peripheral nervous system, affecting approximately 2.9 million people across the world [1]. These lesions of demyelination appear on T1 MPRAGE images as the demyelination reduces the fat content within the white matter of the brain [2]. In this study, we will investigate the use of convolutional neural networks (CNNs) to segment white matter lesions (WML) from normal appearing white matter (NAWM).

This preliminary study is intended to be brief and incomplete, more of an appetiser as opposed to a main course, as the research is ongoing.

Methods

The T1 MPRAGE images for 30 subjects were collected by Dr. Gourab Datta el al [3] and white matter lesion masks were manually identified and labelled by Dr. Gourab Datta using Jim (Xinapse Systems, version 7). PNG images of 32x32 pixels (32mm^2) were then created for each $[i, j]$ slice of the white matter lesion masks that contained at least 24 pixels of tissue identified as lesion tissue. PNGs of the same dimensionality were sliced from the corresponding T1 MPRAGE images. This process was repeated for each subject, over each slice of the WML masks from the superior-inferior plane. In total this produced a set of 6408 brain images with corresponding lesion mask images which were randomly partitioned into a training set of 5408 pairs of images and a validation set of 1000 pairs of images.

The CNN configuration was composed of eight hidden layers; four Conv2D layers with kernel size=3, shapes [8,8,16,16] respectively with strides=2 on each odd layer, and four Conv2DTranspose layers with kernel size=3, shape [16,16,8,8] respectively and strides=2 on each even layer. Each hidden layer's activation function was a rectified linear unit. The output layer was a dense layer containing a single neuron with a sigmoid activation function to create a probabilistic predicted lesion mask. The cost function minimized over the network was binary cross entropy and the optimiser was the ADAM optimizer (Adaptive Moment Estimation). An image of the network topology is available in Appendix 1.

Results

The network was trained over 50 epochs, with loss evaluated as the mean square error (MSE) between the predicted lesion mask and the clinician created "ground truth". At the end of

training, the training set had a MSE = 0.183 and validation MSE = 0.196 and a binary accuracy of 93%. A plot of the generalisation curve is shown below in Fig 1).

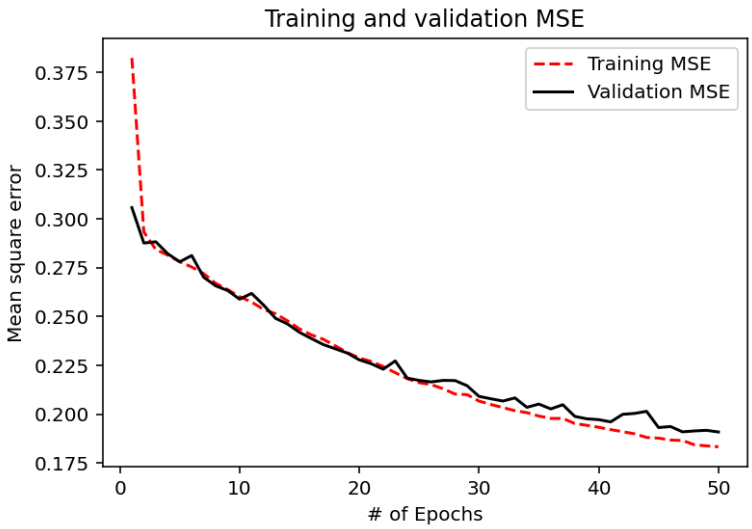


Fig 1) Training and validation MSE over training epochs.

Investigating some of the validation set images, there is some agreement between the predicted lesion maps and the "ground truth" as shown in Fig 2A-C).

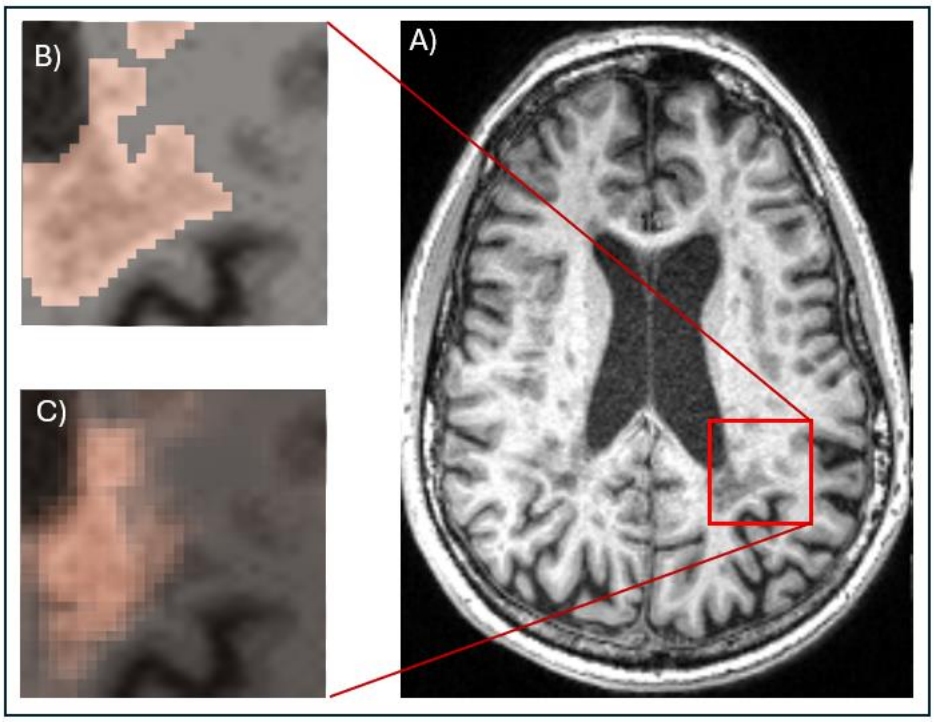


Fig 2A) T1 MPRAGE image. Fig 2B) T1 slice with lesions identified by clinician. Fig 2C) T1 slice with predicted lesions from CNN.

Discussion

This preliminary study shows that CNNs can be used to effectively detect WMLs, though there is still much work that can be done to improve the quality of the boundary of lesions and reduce

the rate of false positives. More modalities could be integrated such as position emission tomography of radioligands specific to activated microglia and reactive astrocytes [3] or T2-weighted FLAIR images which detect water concentration with lesion volumes [3]. Integrating these additional modalities in a multi-modal model would help steer the model away from the misclassification of NAWM as WML, improving model accuracy.

Furthermore, there are situations in which the CNN has identified new lesion masses undetected by the manual segmentation. When manually segmenting lesions, each individual lesion volume on each brain image is manually isolated across the superior-inferior plane. This creates lesion volumes that are not smooth and contiguous, but instead discretised over the other two planes, resulting in inaccurate segmentation. This laborious time-consuming process can leave many small or difficult to detect lesions misclassified as NAWM. Hence a CNN method similar to the one highlighted in this approach may outperform the “ground truth” with sufficient refinement.

Conclusion

There is much to be done to improve the approach outlined in this document, but these preliminary results here show that a CNN model using T1 MPRAGE images is sufficient to identify white matter lesions in a limited capacity.

References

1. https://mstrust.org.uk/information-support/about-ms/how-common-multiple-sclerosis
2. Nair G, Absinta M, Reich DS. Optimized T1-MPRAGE sequence for better visualization of spinal cord multiple sclerosis lesions at 3T. AJNR Am J Neuroradiol. 2013 Nov-Dec;34(11):2215-22. doi: 10.3174/ajnr.A3637. Epub 2013 Jun 13. PMID: 23764721; PMCID: PMC4278432.
3. Datta G, Colasanti A, Kalk N, Owen D, Scott G, Rabiner EA, Gunn RN, Lingford-Hughes A, Malik O, Ciccarelli O, Nicholas R, Nei L, Battaglini M, De Stefano N, Matthews PM. ^{11}C-PBR28 and ^{18}F-PBR111 Detect White Matter Inflammatory Heterogeneity in Multiple Sclerosis. J Nucl Med. 2017 Sep;58(9):1477-1482. doi: 10.2967/jnumed.116.187161. Epub 2017 Mar 16. Erratum in: J Nucl Med. 2020 Sep;61(9):1313. PMID: 28302760.

Appendix 1. Network Topology

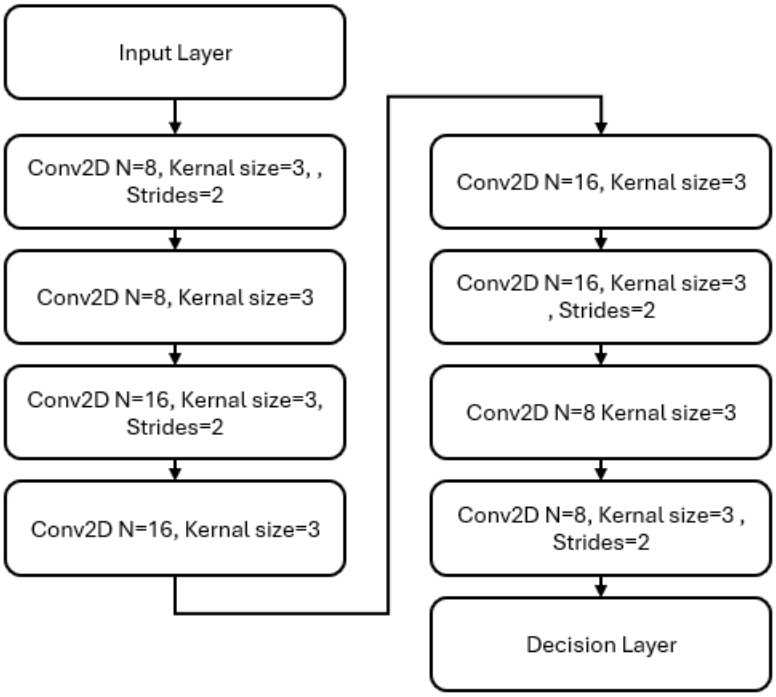


Fig 3) The CNN topology used in this study